CHMB23 · Thermodynamics Ch. 1–3 – Cheat Sheet

heat & work · 1st law · H · C · kinetic theory · entropy · 2nd/3rd law · A, G · engines – built from the 105-question quiz book

Jiasheng Fu

Ideal-gas processes (n mol, constant $C_{V,m}$, $C_{p,m} = C_{V,m} + R$) – ΔU, ΔH, ΔS are state functions: same value for any path between the same endpoints; q, w depend on the path

Process	w (on gas)	q	ΔU	ΔH	ΔS_{sys}	ΔS_{surr} (reservoir)	Notes
Isothermal, reversible	$-nRT\ln(V_f/V_i) = -nRT\ln(p_i/p_f)$	$-w$	0	0	$nR\ln(V_f/V_i)$	$-nR\ln(V_f/V_i) \to \Delta S_{tot} = 0$	max work output; $U = U(T)$
Isothermal, const p_{ex}	$-p_{ex}(V_f - V_i)$	$-w$	0	0	$nR\ln(V_f/V_i)$	$-q/T \to \Delta S_{tot} > 0$	less output than reversible
Free expansion ($p_{ex} = 0$, insulated)	0	0	0	0	$nR\ln(V_f/V_i) > 0$	$0 \to \Delta S_{tot} > 0$	$q = 0$ needs insulation
Isochoric heating	0	$nC_{V,m}\Delta T$	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	$nC_{V,m}\ln(T_f/T_i)$	$-q/T_{surr}$	$q_V = \Delta U$; $\Delta H - \Delta U = nR\Delta T$
Isobaric heating	$-p\Delta V = -nR\Delta T$	$nC_{p,m}\Delta T$	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	$nC_{p,m}\ln(T_f/T_i)$	$-q/T_{surr}$	$q_p = \Delta H$
Reversible adiabatic	$nC_{V,m}\Delta T$	0	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	0 (isentropic)	0	$TV^{\gamma-1}$, pV^γ const; $\gamma = C_p/C_V$
Phase change at T_{trs}, const p	$-p\Delta V$	$\Delta_{trs}H$	$\Delta_{trs}H - p\Delta V$	$\Delta_{trs}H$	$\Delta_{trs}H/T_{trs}$	$-\Delta_{trs}H/T_{trs} \to 0$	freezing $= -\Delta_{fus}H$

Conventions · constants · units 约定与常数

- $\Delta U = q + w$: $q > 0$ heat **into** system, $w > 0$ work **on** system; output $W_{out} = -w$. (Physics books flip the sign of w.)
- $R = 8.314\ \text{J K}^{-1}\text{mol}^{-1} = N_A k_B$; $k_B = 1.380649 \times 10^{-23}\ \text{J K}^{-1}$; $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$. Use k_B with N, R with n; $pV = nRT = Nk_BT$.
- RT at 298.15 K = 2.479 kJ mol⁻¹. $R\ln 2 = 5.76$, $R\ln 1.5 = 3.37$, $R\ln 3 = 9.13$ J K⁻¹ mol⁻¹.
- $T/\text{K} = \theta/^\circ\text{C} + 273.15$; a difference needs no offset. **273.16 K** = water triple point, not the offset.
- $p^\circ = 1$ bar $= 10^5$ Pa (IUPAC) ≠ 1 atm $= 101.325$ kPa $= 1.01325$ bar.
- **SATP** = 298.15 K & 1 bar → $V_m = RT/p = 24.79$ L mol⁻¹. Old STP = 273.15 K & 1 atm (22.41 L mol⁻¹).
- Standard state ($^\circ$ = $^\ominus$) fixes the pressure (gas: pure ideal gas at 1 bar), **not** T: $\Delta_r H^\circ(T)$ exists at any T; 298.15 K is just the usual table value.
- Always: T in K in ratios, ln, q/T, $e^{-\varepsilon/kT}$; kJ → J (×1000); M in kg mol⁻¹.
- **Before any formula ask**: system, surroundings or total? Fixed $T, p/T, V$? Reversible (else invent a rev path for ΔS_{sys})? Ideal gas, closed, pV work only?

Systems · walls · state functions 体系

- Open (matter + energy) · closed (energy only) · isolated (neither). **Diathermic** wall lets heat through (no net flow if T equal). **Adiabatic** wall: $q = 0$; does not by itself block work or matter, and T can change (adiabatic ≠ isothermal).
- Equilibrium: thermal → equal T; mechanical (frictionless movable wall) → $p_1 = p_2$, *not* equal V ($V_1/V_2 = n_1/n_2$ at equal T); chemical → equal μ.
- **State** functions U, H, S, A, G ($\oint dX = 0$, endpoints only). **Path** functions q, w (inexact: write $\delta q, \delta w$); one ΔU splits differently: $q_1 + w_1 = q_2 + w_2$. Cycle: $\Delta U = 0$ but $q_{net} = -w_{net}$ need not be 0.
- **Extensive**: U, H, S, V, total C (J K⁻¹; doubles with n). **Intensive**: $T, p, C_m = C/n$ (J K⁻¹mol⁻¹). Intensive ≠ constant.
- Work = **ordered** motion (force × displacement; also electrical, magnetic). Heat = **random** thermal motion driven by ΔT. Heat is not a substance stored in a body.
- **Reversible** = infinitesimal driving force, $S_{gen} = 0$; an ideal limit (needs infinite time). Quasistatic ≠ reversible: a slow piston with friction still makes entropy.

Work 功

- $\delta w = -p_{ex}\,dV$ – **external** pressure; valid for reversible **and** irreversible. Const p_{ex}: $w = -p_{ex}(V_f - V_i)$.
- Free expansion $p_{ex} = 0 \Rightarrow w = 0$, no work on surroundings. Does **not** imply $q = 0$ unless insulated.
- Rev. isothermal: $-nRT\ln(V_f/V_i) \equiv -nRT\ln(p_1/p_2)$, sign included (Boyle). Expansion ⇒ $w < 0$.
- Reversible = **maximum work output** for fixed endpoints & same reservoir (w_{rev} is the most negative). 1 mol, 300 K, $V \to 2V$: $W_{out} = RT\ln 2 = 1.73$ kJ.
- $W_{out} = \int p_{ex}dV$ = area under the p_{ex}-V curve (rectangle for const p_{ex}); gas-p curve only if reversible. $-nRT\,\Delta V/V_1$ = small-change limit only.

First law · enthalpy · calorimetry 第一定律与焓

- Isolated: $q = w = 0 \Rightarrow \Delta U = 0$ (generally total $E = U + E_{kin} + E_{pot}$ is conserved; open systems add matter terms).
- Perfect gas: $U = U(T)$, $(\partial U/\partial V)_T = T(\partial p/\partial T)_V - p = 0$ ⇒ isothermal $\Delta U = 0$, $q = -w$ (rev **or** irrev). Real gas: U depends on V. Monatomic $U = \frac{3}{2}nRT$.
- $H \equiv U + pV$ (**plus**) $= G + TS$; $dH = dU + p\,dV + V\,dp$; $\Delta H = \Delta U + \Delta(pV)$.
- Const V, no other work: $q_V = \Delta U$ – **not ΔH**: $\Delta H = q_V + V\Delta p$ (ideal gas: $\Delta H - \Delta U = nR\Delta T$).
- Const p, pV work only: $q_p = \Delta H$. With extra (electrical/shaft) work: $\Delta H = q_p + w_{other}$.
- **Bomb calorimeter** (rigid, const V, $w_{pV} = 0$) gives ΔU: $q_{cal} = C_{cal}\Delta T$, $\Delta U_{rxn} \approx -C_{cal}\Delta T$. Open / flame (const p) → ΔH.
- Exothermic $\Delta H < 0$ ($q_p < 0$, $q_{surr} = -q_{sys} > 0$); endothermic $\Delta H > 0$ (melting, boiling). Reverse reaction flips sign. Exothermic ⇏ no work, ⇏ $\Delta U = 0$.
- Gas reactions: $\Delta_r H = \Delta_r U + \Delta\nu_g RT$ (ideal, same T, condensed pV ignored). $A(g) \to 2B(g)$: $\Delta_r H - \Delta_r U = RT$. Equal only if $\Delta\nu_g = 0$.

Heat capacity · equipartition · Kirchhoff 热容

- $C_V \equiv (\partial U/\partial T)_V$, $C_p \equiv (\partial H/\partial T)_p$ (subscript is part of the definition); $\Delta U = \int C_V dT$, $\Delta H = \int C_p dT$; C may vary with T.
- Ideal gas: $C_p - C_V = nR$ ($C_{p,m} - C_{V,m} = R$) ⇒ $C_p > C_V$ (const-p heating also does expansion work). General: $C_p - C_V = TV\alpha^2/\kappa_T \geq 0$.
- Equipartition: $\frac{1}{2}k_BT$ per quadratic term. Translation (3 terms): $U_m = \frac{3}{2}RT$; + linear rotation (2) $+RT$; nonlinear (3) $+\frac{3}{2}RT$. $C_{V,m}$: monatomic $\frac{3}{2}R$ ($C_{p,m} = \frac{5}{2}R$, $\gamma = \frac{5}{3}$), linear $\frac{5}{2}R$, nonlinear $3R$ (no vib).
- Classical only: **fails when** $kT \gg \Delta\varepsilon$ – vibrations frozen out at room T. $\langle\varepsilon_{vib}\rangle = \frac{1}{2}\hbar\omega + \hbar\omega/(e^x - 1)$, $x = \hbar\omega/kT$; zero-point energy adds no C.
- **Kirchhoff**: $\Delta_r H^\circ(T') = \Delta_r H^\circ(T) + \int_T^{T'} \Delta_r C_p^\circ dT \approx \Delta_r H^\circ(T) + \Delta_r C_p^\circ(T' - T)$. Uses C_p (not C_V) and a **plus**; linear form needs const $\Delta_r C_p = \sum \nu C_{p,m}$ (products +, reactants −); across a phase change add the latent heat.
- Adiabatic: $q = 0$, $\Delta U = w = nC_{V,m}\Delta T$ (compression heats, expansion cools).

Kinetic theory · gases · Boltzmann 动理论

- Kinetic model: ceaseless random motion (classical), point molecules (size ≪ separation), perfectly elastic collisions (total KE conserved, not each speed), no interactions between collisions. **NOT**: strong long-range attraction.
- $pV = \frac{1}{3}Nm\langle v^2\rangle$ and $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_BT \Rightarrow pV = nRT$.
- $v_{rms} = \sqrt{3RT/M} > \bar{v} = \sqrt{8RT/\pi M} > v_{mp} = \sqrt{2RT/M}$ (1.225 : 1.128 : 1); all $\propto \sqrt{T/M}$. N_2 at 300 K: $v_{rms} \approx 517$ m s⁻¹. $3RT/M$ alone is v_{rms}^2.
- Maxwell $f(v) \propto v^2 e^{-Mv^2/2RT}$: heating → peak moves **right**, curve **broader & lower**, area stays 1. Cooling → left, narrow, tall.
- Boltzmann $N_j/N_i = (g_j/g_i)\,e^{-(\varepsilon_j-\varepsilon_i)/kT}$: $T \uparrow \Rightarrow$ upper/lower ratio ↑ (higher states fill at the expense of lower).
- Partial pressure $p_J \equiv x_J p$ (real or ideal); $\sum p_J = p$; ideal: $p_J = n_J RT/V$ with the **whole** V.
- $Z = pV_m/RT = 1 + B/V_m + C/V_m^2 + \ldots$; $p \to 0$ at fixed T: molecules far apart, forces & own volume negligible, $Z \to 1$ (speeds still set by T).
- Perfect gas: zero interactions, no excluded volume. **Ideal mixture**: unlike-pair = like-pair interactions (need not be zero): $\Delta_{mix}H = \Delta_{mix}V = 0$, $\Delta_{mix}S = -nR\sum x_i \ln x_i$, $\mu_i = \mu_i^* + RT\ln x_i$.
- Box levels $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2) \propto V^{-2/3}$: bigger V → denser levels → more microstates; $q_{trans} = V/\Lambda^3$, $\Lambda = h/\sqrt{2\pi mk_BT} \Rightarrow \Delta S = nR\ln(V_f/V_i)$.

Entropy & the second law 熵与第二定律

- Definition $dS = \delta q_{rev}/T$ (state function, J K⁻¹). Irreversible process → invent a reversible path between the same endpoints. Not dH/T (only rev, const p), not $\delta q/T$, never work.
- **Clausius** $dS \geq \delta q/T$, equality iff reversible. Isolated: $\Delta S = S_{gen} \geq 0$ (> 0 for spontaneous irreversible change).
- Balance $\Delta S = \int \delta q/T_b + S_{gen}$: work carries **no** entropy, but dissipation does make it (stirring an insulated liquid: $q = 0$, $\Delta S = S_{gen} > 0$). Reversible adiabatic ⇒ isentropic.
- Why $1/T$: the same q gives a bigger ΔS in a cold body (100 J: 1 J K⁻¹ at 100 K, 0.1 at 1000 K; 0.4 at 250 K vs 0.2 at 500 K).
- Statistical $S = k_B \ln W$, W = number of microstates of the macrostate; $W = 1 \Rightarrow S = 0$. Gibbs $S = -k_B\sum p_i \ln p_i$. More volume / energy / freedom (s → l → g) ⇒ $W \uparrow \Rightarrow S \uparrow$.
- Three levels: $\Delta S_{sys} = \int \delta q_{rev}/T$; $\Delta S_{surr} = -q_{sys}/T_{surr}$ (reservoir: **same rule** whether the system is rev or irrev, but q_{sys} depends on the path); $\Delta S_{tot} = \Delta S_{sys} + \Delta S_{surr} \geq 0$.
- Heat Q flowing hot → cold: $\Delta S_{tot} = Q(1/T_c - 1/T_h) > 0$.
- Judge the **total**: $\Delta S_{sys} < 0$ is allowed if $\Delta S_{surr} > |\Delta S_{sys}|$ (water freezing below 0 °C: $\Delta S_{tot} \approx L(1/T - 1/T_m) > 0$).

ΔS recipes 熵变计算

- Ideal gas: $\Delta S = nR\ln\frac{V_f}{V_i} + nC_{V,m}\ln\frac{T_f}{T_i} = -nR\ln\frac{p_f}{p_i} + nC_{p,m}\ln\frac{T_f}{T_i}$ – any path, free expansion too. 1 mol: V doubled +5.76 J K⁻¹, halved −5.76.
- Heating: $\Delta S = \int \frac{C_p}{T}dT$ (const p), C_V at const V. **Not** $C\Delta T$ (= q), not $\Delta T/T_i$; molar $C \times n$.
- Transition at T_{trs} (reversible, const p): $\Delta_{trs}S = \Delta_{trs}H/T_{trs}$ (divide; K). Freezing = −fusion: 1 mol water at 273.15 K: $-6010/273.15 = -22.0$ J K⁻¹ (surr +22.0, total 0).
- $\Delta_{vap}S \approx 85$ J K⁻¹mol⁻¹ (Trouton; 80–90) ≫ $\Delta_{fus}S$ (8–25): gas has ~10³× volume and full translational freedom. $S^\circ_{m,g} > S^\circ_{m,l}$. Both endothermic; fusion breaks only intermolecular forces.
- Source hotter than T_{trs}: $\Delta S_{tot} = \Delta H_{trs}(1/T_{trs} - 1/T_h) > 0$.
- Reaction: $\Delta_r S^\circ = \sum \nu S^\circ_m(\text{prod}) - \sum \nu S^\circ_m(\text{react})$; S°_m are absolute (elements ≠ 0, unlike $\Delta_f H^\circ$).
- $\Delta S_{surr} = -\Delta H_{sys}/T$ (const T, p): exothermic warms & raises S_{surr}. Body 100 W × 86 400 s = 8.64 MJ ⇒ $\Delta S_{surr} = 8.64 \times 10^6/293.15 = +29.5$ kJ K⁻¹.

Third law · residual entropy 第三定律

- Perfect crystal: $S \to 0$ as $T \to 0$ (unique ground state, $W = 1$) → absolute entropies.
- $S^\circ_m(T) = S_{res} + \sum_{\text{phases}} \int \frac{C_{p,m}}{T}dT + \sum \frac{\Delta_{vap}H}{T_{trs}}$ = area under C_p/T vs T plus the jumps.
- $T \to 0$: $C \to 0$. Debye $C_{V,m} \approx aT^3$ ($T \ll \Theta_D$; metals add a $\propto T$ term) ⇒ $S = \int_0^T aT'^2 dT' = \frac{1}{3}aT^3 = \frac{1}{3}C$ (**not** $\frac{1}{4}aT^4, C, 3C$). $C = aT^n \Rightarrow S = C/n$.
- Residual $S = k_B \ln W_0$. **Ice**: proton disorder (O–H vs O···H) $W \approx (3/2)^N \Rightarrow R\ln 1.5 = 3.4$ J K⁻¹mol⁻¹ (not isotopes, electrons, translation). **CO/OC**: $W = 2^N \Rightarrow R\ln 2 = 5.76$ (real ≈ 4–5). 3 orientations → $R\ln 3 = 9.13$.
- $C \to 0$ does not prove $S(0) = 0$: glass, solid solutions, ice, CO keep S_{res}. "Solid" ≠ one microstate.

Free energy · spontaneity · max work 自由能

- $A \equiv U - TS$, $G \equiv H - TS = A + pV$. Fundamental: $dU = TdS - pdV$, $dH = TdS + Vdp$, $dA = -SdT - pdV$, $dG = -SdT + Vdp$.
- Criteria (closed, pV work only): fixed T, V: $\Delta A = -T\Delta S_{tot} \leq 0$; fixed T, p: $\Delta G = -T\Delta S_{tot} \leq 0$ (< 0 spontaneous, = 0 equilibrium). $\Delta H < 0$ is a test only at fixed S, p.
- Why: $\Delta S_{surr} = -\Delta H/T \Rightarrow \Delta G = \Delta H - T\Delta S_{sys} = -T(\Delta S_{sys} + \Delta S_{surr})$.
- **Max work (sign!)**: fixed T: $W_{out,max} = -\Delta A$ (total work; $w_{on,rev} = \Delta A$). Fixed T, p: $W_{non\text{-}pV,max} = -\Delta G$ (e.g. cell work). ΔG is not expansion work.
- $\Delta G - \Delta A = \Delta(pV) = nR\Delta T$ (ideal gas): equal iff isothermal (fixed n) – not "any process".
- $\Delta_r G = \Delta_r G^\circ + RT\ln Q$: direction from $\Delta_r G$, not $\Delta_r G^\circ$.
- $\Delta G = \Delta H - T\Delta S$: (−,+) always; (+,−) never; (−,−) low T; (+,+) high T, $T > \Delta H/\Delta S$.
- Spontaneous = tends to happen with no work input; says nothing about rate (diamond → graphite). Endothermic can be spontaneous (ice melts at 25 °C). Spontaneous ⇒ S_{tot} **increases**.

Engines · refrigerators · heat pumps 热机与制冷

- Engine $\eta = W/Q_h = 1 - Q_c/Q_h \leq \eta_C = 1 - T_c/T_h$ (K!). Carnot: $Q_h/T_h = Q_c/T_c$, $W = Q_h - Q_c$.
- 473.15 / 293.15 K → $\eta = 0.380 \approx 38\%$. **Traps**: 90% = $(200 - 20)/200$ in °C; 61.9% = T_c/T_h = fraction of heat rejected.
- $\eta < 1$ unless $T_c = 0$ K. Hotter source or colder sink raises η: $\partial\eta/\partial T_h = T_c/T_h^2 > 0$, $\partial\eta/\partial T_c = -1/T_h < 0$. The **ratio** matters: (600, 300) K → 0.50, (900, 600) K → 0.333.
- **Kelvin-Planck**: no cycle turns heat wholly into work (one reservoir: $\Delta S_{tot} = -Q_h/T_h < 0$). One isothermal step can ($q = -w$), a full cycle cannot. Two reservoirs: $-Q_h/T_h + Q_c/T_c \geq 0$.
- Fridge $COP_{cool} = Q_c/W = T_c/(T_h - T_c)$; heat pump $COP_{warm} = Q_h/W = T_h/(T_h - T_c) = COP_{cool} + 1$. Both can exceed 1; smaller ΔT → larger COP.
- A fridge obeys the 2nd law: cold → hot is not the sole effect (work goes in); $\Delta S_{tot} = Q_h/T_h - Q_c/T_c \geq 0$.

Yes/No quiz facts 判断题速查

Heat & work (27)

- Y Vacuum expansion: $w = 0$. N It does work on the surroundings.
- N Perfect-gas U depends on V (const T). Y Isothermal: $\Delta U = 0$.
- Y Reversible ⇒ max work output (fixed endpoints).
- N U is a path function. N w is a state function. Y Cycle: $\Delta U = 0$.
- N $q_V = \Delta H$ (it is ΔU). Y $q_p = \Delta H$ (const p, pV work only).
- N $C_p < C_V$. N Total C_V is intensive. N C_m is extensive.
- Y Exothermic ⇒ $\Delta H < 0$. N Endothermic ⇒ $\Delta H < 0$.
- N Reversible process finished in finite time. Y 1st law ≡ isolated U constant.
- Y Heat in: $q > 0$. N Standard p = 1 atm. N Standard state needs 298.15 K.
- Y $C_V = (\partial U/\partial T)_V$. N $\Delta H = \Delta U$ whatever Δn_g.
- Y Every pure substance: $C \to 0$ as $T \to 0$.
- Y $dw = -p_{ex}dV$ for rev and irrev. Y Work = area under the p-V curve.
- N A diathermic wall blocks heat. N $H = U - pV$.

Entropy · 2nd law · free energy (26)

- Y Isolated, spontaneous: $\Delta S \geq 0$. N Carnot $\eta = 100\%$ with $T_c > 0$.
- N Reservoir ΔS_{surr} depends on rev vs irrev.
- Y Isothermal $\Delta S = nR\ln(V_f/V_i)$. N Isothermal compression raises S_{gas}.
- N Same q at low T → smaller ΔS. Y $W = 1 \Rightarrow S = 0$.
- Y Ice has residual S (H-bond disorder). N Solid CO has $S = 0$ at 0 K.
- Y Fixed T, V: $\Delta A < 0$ is the test. Y Fixed T, p: $\Delta G < 0 \Rightarrow$ spontaneous.
- N S°_m(gas) < S°_m(liquid). Y At T_{trs}: $\Delta S_{sys} = -\Delta S_{surr}$.
- N A cycle turns heat fully into work, no waste heat.
- Y Debye: $C_{p,m} \to 0$ as $T \to 0$. Y Debye: $S_m = \frac{1}{3}C_{p,m}$.
- Y $\Delta S_{surr} = -\Delta H/T$ (const p). Y Spontaneous with $\Delta S_{sys} < 0$.
- N Work changes S directly via disorderly motion.
- Y $S^\circ(T) = a$ area under C_p/T vs T (+ transitions).
- Y $k = R/N_A$. Y Const V: $\Delta S = \int C_V/T\,dT$.
- N A fridge violates the 2nd law. Y Hotter source + colder sink ⇒ $\eta \uparrow$.
- Y Expansion → translational levels closer. N $\Delta G = \Delta A$ for any ideal-gas process.

Multiple-choice traps 选择题陷阱

- **Two right options**: $-nRT\ln(P_1/P_2) \equiv -nRT\ln(V_2/V_1)$; $H = G + TS$ is also true; at $p_{ex} = 0$, $+p_{ex}\Delta V$ is also 0.
- "Max work = ΔA / ΔG": positive output is $-\Delta A$ (total) / $-\Delta G$ (non-pV).
- Ideal mixture = equal interactions ("non-zero" not required); perfect gas = no interactions.
- °C in a ratio; $T_c/T_h \neq \eta$; COP forms swapped or inverted (< 1).
- $\int C\,dT$ vs $\int C/T\,dT$; $C\Delta T$ is heat, not ΔS; ln ratio upside down; × vs ÷ T_{trs}.
- Forgot ×1000 (−6.01 J K⁻¹) or ÷T (8.64 MJ K⁻¹); sign of surroundings (gains heat ⇒ +).
- Speeds swapped ($\sqrt{8/\pi}$ mean, $\sqrt{2}$ m.p., $\sqrt{3}$ rms); missing $\sqrt{}$.
- Kirchhoff with C_V or a minus sign; $\Delta_r S^\circ$ = products − reactants (not ΔH°/298.15).
- $A = U - TS$ (not $H - TS = G$; $U + pV = H$; $H + TS$ meaningless). C_V vs C_p subscripts.
- $\Delta S = 0 \Rightarrow$ reversible adiabatic, not isothermal; $\Delta S_{surr} = 0$ only if no heat exchanged.
- Spontaneous ≠ fast; spontaneous never "decreases total S"; exothermic not required.